SAMPLE

Earth's Sky: Supermoons, Blue Moons, and Blood Moons

Special Lunar Events Explained

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Introduction

Three titles have collected on the Moon in the past half-century, one borrowed from astrology, one from a misheard word, and one from the color the Moon turns when the Earth stands between it and the Sun. The supermoon, the blue moon, and the blood moon are not new phenomena; the Moon has been doing all three since before there were names for them. What is new is the naming, and with it a genuinely modern fact: the Moon now makes headlines, and millions of people step outside on a scheduled evening to look up at events their grandparents watched without a label. This book is about those three titles, what each one actually means, how each got its name, and what the sky is really doing when the news tells everyone to go outside.

The three titles are worth one book because they belong to one subject: the full Moon at the moments it is special. A supermoon is a full Moon near perigee, the closest point of the Moon's orbit, when the disk runs some fourteen percent larger in area than at its smallest, and shines noticeably brighter. A blue moon is a matter of counting rather than appearance, a month's second full Moon, or by the older reckoning the season's third full Moon when the season holds four, a curiosity of the arithmetic of the Moon's cycle against ours. A blood moon is a total lunar eclipse, the full Moon sliding into the Earth's shadow and, instead of vanishing, glowing a deep copper red with the light of all the world's sunrises and sunsets at once. Two of the three are geometry, and one is bookkeeping, and all three are the same familiar satellite caught at its extremes.

The book keeps to the series' discipline about evidence. Every quantity in it, the sizes, distances, periods, and angles, comes from the standard, well-established account of lunar motion and eclipses, and where a fact is a matter of folklore or record rather than measurement, the phrasing says so. The history of the names is told as history: the supermoon was coined in 1979 by an astrologer and adopted by the press decades later, the blue moon's meaning shifted through centuries of misquotation, and the blood moon is the newest coat of paint on the oldest observed phenomenon in astronomy. The reader will find no invented drama here, only the genuine article, which is dramatic enough: a sky in which the same face that farmers planted by turns, on schedule, huge and red in the dark.

The road map follows the series' pattern, so the reader can steer by chapter number alone. The first chapter gives the discovery history, from Babylonian eclipse records through the coinages and media waves that gave the phenomena their modern names. The second and third chapters set out the fundamentals, the Moon's orbit and the three definitions, and the catalog of what each event looks like. The fourth through sixth chapters go deeper into the science, the observation, and the signatures by which the three events are recognized. The seventh places them in context, in calendars, folklore, tides, and the news; the eighth through eleventh report the modern account, the research, the discoveries, and the open questions. The twelfth looks forward, the thirteenth collects the trivia, and the fourteenth delivers the closing argument for why any of it matters. The reader who comes this far will know, on the next newsworthy night, exactly what the Moon is doing and why, and will enjoy the crowd around them a great deal more for it.

A note on the book's scale, since a reader planning their evenings deserves it. Of the three events, the blood moon is the great one: it is visible from the entire night side of the Earth when it happens, it unfolds for hours, and it requires no knowledge at all to be struck by. The supermoon is the sociable one, a scheduled appointment that fills parks and rooftops, best enjoyed with a camera and a foreground and a small dose of realism about what the eye can actually see. The blue moon is the domestic one, an event that happens in the calendar and is celebrated mostly in the telling, and its observation consists of knowing, on an ordinary night, that the Moon overhead is doing something arithmetically rare. The book's promise is that all three kinds of evening are improved by understanding, and that none of the three requires anything of the reader but the glance upward that the title was invented to provoke. The chapters that follow keep the series' plain style: no equations beyond arithmetic, every number one the reader can check, and the sky explained as the watchable, schedulable, generous thing that four centuries of astronomy have shown it to be.

Chapter 1: The Discovery of Supermoons, Blue Moons, and Blood Moons

The phenomena are as old as the sky; the discoveries about them came in two waves, an ancient one that learned to predict the events, and a modern one that gave them their names. This chapter tells both stories, and the long quiet stretch between them in which the Moon did its work without a headline.

The blood moon has the oldest history by thousands of years, because a total lunar eclipse is unmissable and

alarming. The Babylonians recorded them systematically, and by the last centuries before the common era their astronomers had found the pattern of eclipses, the roughly eighteen-year cycle after which eclipses of similar character return, a discovery that turned the blood-red Moon from omen into schedule. Greek natural philosophers drew the correct conclusion from the Moon's behavior in shadow, that the Earth is a sphere, and the round edge of our planet, printed on the Moon during every eclipse, has served as the standing proof ever since. Through the medieval period, eclipse prediction remained a live practice in the Islamic world, in monasteries, and at the courts of China, and by the time Tycho and Kepler refined the lunar tables at the close of the sixteenth century, the blood moon was the most predictable of all sky events.

The blue moon's history is the history of an idiom rather than an event. The phrase once meant an absurdity, the kind of thing that never happens, and its migration toward astronomy came through a chain of misreadings and simplifications over several centuries. The decisive step was an article in an American farm almanac tradition in the nineteen-forties that defined the blue moon as the season's third full Moon in seasons holding four, a rule that kept the names of the seasonal Moons in their proper places; a later simplification, born of a summary of that rule in the nineteen-eighties, recast it as a month's second full Moon, and the newer, simpler definition captured the public imagination and now owns the phrase. Nothing physical was discovered in either step; what was discovered was a piece of arithmetic charm, and it proved sturdy enough to become a permanent fixture of the language.

The supermoon's discovery history is shorter and stranger, because the word is younger than most of its readers. It was coined in 1979 by an astrologer, Richard Nolle, in a piece of writing about the Moon near perigee, with a deliberately loose definition tying unusual closeness of full or new Moon to terrestrial effects. The astronomical establishment pointed out, correctly, that nothing about perigee syzygy was new, that the Moon has always swung close and far, and that the term had no scientific standing; the press adopted it anyway in the twenty-teens, and by the middle of that decade every perigee full Moon was a scheduled news event, with photographers, crowds, and even a piece of accepted practice, the handful of days when everyone with a camera looks up. The adoption of an astrologer's coinage by the world's newsrooms, over the objections of the astronomers, is one of the odd little sociologies of modern science communication, and this book reports it as a fact of media history rather than a quarrel to be re-fought.

Two threads run through all three histories, and the chapters that follow pick them up. The first is that the phenomena were fully understood long before they were famous: Kepler's eclipse tables, the lunar distance measures of the eighteenth century, and the orbit theory of the nineteenth made the Moon the best-mapped object in the sky while the blue moon was still an idiom and the supermoon was not yet a word. The second is that the modern fame of all three has been genuinely good for observation, whatever the purists think; the crowds at a supermoon or an eclipse are standing outside, at night, looking up, and more people have now watched the Earth's shadow crawl across the Moon than at any time in history. The discovery story of these three titles ends where the observer's own story begins, on the next newsworthy evening, with the Moon overhead behaving exactly as four centuries of prediction promised.

One more strand of the discovery history deserves its paragraph, because it explains why the three titles are famous now rather than a century ago: the events became photographable and shareable at exactly the moment the Moon's fame revived. An eclipse watched alone is a memory; an eclipse photographed is an argument, and the spread of decent cameras through the population, followed by the networks that move pictures in seconds, turned every blood moon into a global album and every supermoon into a wall of skyline shots, honestly made and endlessly compared. The blue moon never needed the camera; it is a calendar event, and its spread was purely verbal, riding the idiom that had been waiting for centuries. The historians of the subject point out that all three titles completed their journey to the public in the twenty-teens, the same half-decade, and the coincidence is not a coincidence: the same networks that carried the pictures carried the words, and the sky, which had been receding from daily life for two industrial centuries, returned to the feed. That return is the discovery history's final entry, and the chapters that follow describe what the returning public actually found waiting, which was the oldest, nearest, and best-explained object in the sky, wearing its three new hats.